Algorithm Comparison Report

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Scope

This report compares the cryptographic algorithms in the benchmark using the mobile datasets from unplugged_run/d1 and unplugged_run/d2.

The comparison is focused on three criteria:

1. Average total time
2. Average energy consumption
3. Composite trade-off score

The analysis uses the already processed notebook outputs, where lower values are better for all three criteria.

1. Comparison by Average Time

Rank	Algorithm	Mean time (ms)	Median (ms)
1	AES-GCM	6.3150	4.5796
2	ChaCha20	9.6171	5.1231
3	AES	12.1958	5.2467
4	ASCON	12.8408	8.6229
5	Xoodyak	16.2028	8.8625
6	RSA-Hybrid	37.9185	31.1956
7	ElGamal	38.4988	31.3004
8	GIFT-COFB	52.1101	25.1930
9	Grain-128AEAD	90.6167	33.2195
10	Elephant	225.9220	80.0766

AES-GCM is the fastest algorithm on average. ChaCha20 is the second-best in speed and remains clearly faster than most of the remaining algorithms.

2. Comparison by Average Energy

Rank	Algorithm	Mean energy (J)	Median (J)
1	ElGamal	0.000193	0.000031
2	RSA-Hybrid	0.000213	0.000031
3	AES-GCM	0.000458	0.000031
4	ASCON	0.000521	0.000031

Rank	Algorithm	Mean energy (J)	Median (J)
5	Xoodyak	0.000613	0.000031
6	GIFT-COFB	0.000766	0.000029
7	ChaCha20	0.000768	0.000031
8	AES	0.001488	0.000031
9	Grain-128AEAD	0.001563	0.000031
10	Elephant	0.002201	0.000037

ElGamal has the lowest mean energy, followed closely by RSA-Hybrid. However, these methods are much slower than AES-GCM, so they are not the best balanced choices.

3. Composite Trade-Off Score

The notebook computes a normalized score using:

- time_norm = min-max normalized total time
- energy_norm = min-max normalized energy
- score = 0.5 * time_norm + 0.5 * energy_norm

Lower scores are better.

Rank	Algorithm	Mean score	Mean time (ms)	Mean energy (J)
1	AES-GCM	0.0207	6.3150	0.000458
2	ElGamal	0.0242	38.4988	0.000193
3	RSA-Hybrid	0.0249	37.9185	0.000213
4	ASCON	0.0275	12.8408	0.000521
5	Xoodyak	0.0341	16.2028	0.000613
6	ChaCha20	0.0386	9.6171	0.000768
7	GIFT-COFB	0.0613	52.1101	0.000766
8	AES	0.0774	12.1958	0.001488
9	Grain-128AEAD	0.1234	90.6167	0.001563
10	Elephant	0.2293	225.9220	0.002201

AES-GCM is the best overall algorithm under the balanced trade-off score. It combines the best runtime with a competitive energy profile, which makes it the strongest general-purpose choice in this benchmark.

Main Conclusion

If the goal is to choose one algorithm from the benchmark, AES-GCM is the best overall option.

- It is the fastest algorithm on average.
- It is competitive in energy consumption, even if it is not the absolute lowest.
- It also ranks first in the combined score that balances time and energy equally.

If the goal is pure energy minimization, ElGamal is the best. If the goal is pure speed, AES-GCM is the best. For a practical mobile benchmark recommendation, AES-GCM is the most balanced choice.

Figures Used in the Notebook

The notebook already includes plots that support this comparison:

- time_trends_by_algorithm_device.png
- energy_memory_trends.png
- algorithm_comparison_by_device.png
- device_comparison_by_algorithm.png
- round_level_variability_total_time.png

Exported Tables Used Here

- summary_rankings.csv
- tradeoff_ranking.csv